

◦ Aim:

- To determine the percentage composition of the constituents in a binary solution using Abbe's refractometer.

◦ Apparatus:

- Binary solution of various compositions, Abbe's Refractometer, etc.

◦ Theory:

- Refractive index of a binary mixture approximately depends on the composition of a binary solution. Therefore this refractive index measurements can be used to determine the percentage composition of a binary solution. Standard solutions are prepared with the two constituents present in the unknown mixture. The refractive index of the standard mixtures and that of the unknown mixtures are determined using Abbe's refractometer. A graph is obtained by plotting refractive index against percentage composition of one of the constituents. Knowing the refractive index of the unknown mixture, the percentage composition can be obtained from the graph. The principle of Abbe's refractometer is based on total internal reflection.

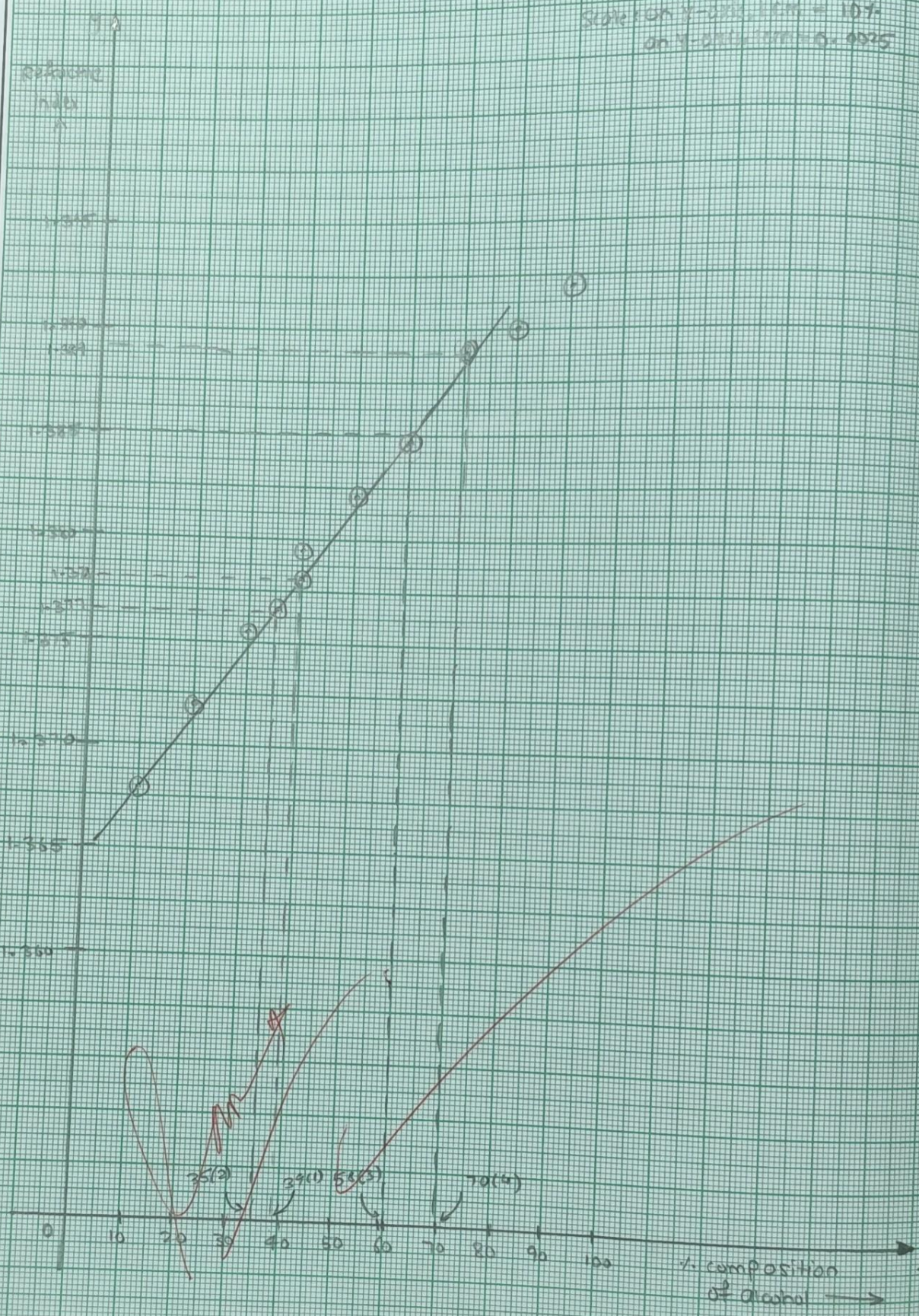
◦ Procedure:

- Prepare mixtures of different compositions of constituents A (water) and B (ethanol) by pipetting different volumes of A and B. Place the refractometer in such a way that light from the source is reflected by the mirror into the split prism. Adjust the mirror for max. illumination. Open the prism box and clean the diagonal surface on the prism with acetone using a cotton swab.

° Observation:

Sr. No.	% composition (water)	% composition (ethanol)	Refractive Index
1	10	90	1.392
2	20	80	1.390
3	30	70	1.389
4	40	60	1.387
5	50	50	1.383
6	60	40	1.381
7	70	30	1.376
8	80	20	1.372
9	90	10	1.367
10	Unknown 1 →		1.378
11	Unknown 2 →		1.377
12	Unknown 3 →		1.385
13	Unknown 4 →		1.389

Scale: on y-axis, 1 cm = 10°
 on x-axis, 1 cm = 0.0025



- when the prism surface is dried, place a drop of the liquid to form a thin film upon the ground surface of the lower prism. Close the prism box immediately. Adjust the mirror to get good illumination. By turning the milled ring at the base of the telescope, a position is found at which the colored band is eliminated and field of view being partly dark and partly light with a sharp dividing line is obtained. Adjust the eyepiece to the position where line of demarcation coincides at the center of the crosswire. At this position, read out the refractive index of the sample through another eyepiece. Similarly, find the refractive index values of other standard and unknown solutions too.
- Plot a graph of refractive index (y-axis) against % composition of ethanol to get a straight line. From the graph, the composition of the unknown solution can be obtained.
- o Result:
- Composition of mixtures are:
 - (i) solution 1 is: 39% ethanol, 61% water.
 - (ii) solution 2 is: 35% ethanol, 65% water.
 - (iii) solution 3 is: 58% ethanol, 42% water.
 - (iv) solution 4 is: 70% ethanol, 30% water.